

Masoud SADRNEZHAAD

PhD Student at Linköping University

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I am a PhD candidate in the Complex Software-Intensive Systems Research Group at Linköping University. My research focuses on the intersection of cyber-physical systems modeling and LLMs. Before my PhD, over a 12-year span, I held roles as a web developer, technical lead, software architect, recommender systems researcher, programming instructor, and Linux server administrator.

Competencies: Software Engineering Agentic AI Recommender Systems Modelica Web Development Linux

EDUCATION

- Oct 2024 Now** **Doctor of Philosophy - PhD in Computer Science and Engineering**
Location: Linköping University, Linköping, Sweden
‣ **Research Area:** LLM-based diagnosis and repair of Modelica models
‣ **Advisor:** Dr. Daniel Varro
‣ **Activities and Societies:** Active committee member at Navitas.se, managing social media posts and event posters for the largest student association focused on sustainable development.
- Sep 2019 Oct 2022** **Master of Science - MSc in Information Technology (Subdiscipline: Enterprise Architecture)**
Location: Shahid Beheshti University, Tehran, Iran (Ranked 741th university in the world by QS 2026)
‣ **GPA:** 19.46 / 20 (1st Rank)
‣ **Thesis Title:** Using empirical data to support technology selection in software architecture decision-making
‣ **Advisor:** Dr. Sadeqh Aliakbary
‣ **Activities and Societies:** Coordinated executive-team activities and workshop lecturer communications for the Third National Conference on Enterprise Architecture. conference team page
- Sep 2012 Jun 2018** **Bachelor of Science - BSc in Computer Engineering (Subdiscipline: Software Engineering)**
Location: Sharif University of Technology, Tehran, Iran (Ranked 375th university in the world by QS 2026)
‣ **Activities and Societies:** Initiated the Free Software Users Group, contributed to Rayanesh magazine, and supported Sharif AI Challenge 2015 technical infrastructure.

PUBLICATIONS

- Sadrnezhaad, M., Hernández López, J.A., Mårtensson, T., Varró, D.: *Generative AI in Simulation-Based Test Environments for Large-Scale Cyber-Physical Systems: An Industrial Study*. In: *Product-Focused Software Process Improvement (PROFES 2025)*, Salerno, Italy, December 1–3, 2025. Lecture Notes in Computer Science, vol. 16361, pp. 1–17. Springer, (2026).
10.1007/978-3-032-12089-2_13 (Full research paper; CORE B; 37% acceptance rate.)

TEACHING EXPERIENCE

- **Accredited University Courses:**
‣ I contributed as a lab assistant, supporting course delivery through course planning, grading exams and assignments, assisting students in problem-solving sessions, and managing online course forums.

University	Course Title	Level	Examiner	Semester(s)
Linköping University	Methods and Tools for Large Distributed Projects	MSc	Dr. August Ernstsson	Winter 2026, Winter 2025
	Concurrent Programming and Operating Systems	BSc		Winter 2026, Winter 2025
Shahid Beheshti University	Advanced Software Engineering	MSc	Dr. Fereidoon Shams Aliee	Fall 2021
Sharif University of Technology	System Analysis and Design	BSc	Seyed Ebrahim Abtahi	Fall 2017, Spring 2017, Fall 2016
			Dr. Mehraneh Shantiaei	Spring 2017
	Advanced Programming in Java	BSc	Dr. Salman Marvasti	Fall 2015
	Technical Presentation	BSc	Dr. Ali M. A. Hemmatyar	Fall 2014

- **Thesis Supervision:**
‣ **Amin, M. and Rosengren, T.** *AI Agent for Asynchronous Interaction in Telecom Configuration Tool*. Master's thesis, Linköping University, Department of Computer and Information Science, Datateknik, 30 ECTS, 2026. (scheduled) **Supervisors:** M. Sadrnezhaad (Internal), I. Persson (External). **Examiner:** D. Varró. **External Commissioner:** Ericsson AB.
‣ **Paulsson, M.** *Improving Quality in AI-Generated Code through Prompt Engineering – An Evaluation of the Impact of Prompting Techniques on Software Quality in Code Generated by Large Language Models*. Master's thesis, Linköping University, Department of Computer and Information Science, Datateknik, 30 ECTS, 2025. Available online. **Supervisors:** M. Sadrnezhaad (Internal), T. Mårtensson (External). **Examiner:** D. Varró. **External Commissioner:** SAAB AB.

External Extracurricular Courses:

- Taught 200+ students with high satisfaction rates: 99% believe the instructor knows the subject well, 98% find the course well-organized, 97% enjoy the current content and respectful, friendly style, and 91% appreciate prepared class plans.
- Developed courseware that includes announcements, resources, topics, calendars, assignments, and code samples.

Course Title	Organizer	Semesters
Python Programming	Sharif University of Technology - Sharif Rayan Academy	Spring 2019, Winter 2020
	University of Tehran - Cultural Education and Development Centre	Fall 2018
	Saipa Training Center	Summer 2019
	MapsHR Company	Fall 2021
	Shahid Beheshti University - Culture Home Center	Summer 2017, Fall 2017, Winter 2018, Summer 2018, Fall 2018, Winter 2019, Summer 2019
Front-end Web Development		Summer 2016, Fall 2018, Winter 2019, Summer 2019
Linux System Administration		Fall 2016
Web Scraping in Python		Winter 2018

Public Engagement & Invited Talks

- Dec 2020 **Animating and Simulating Business Processes to Redesign Them**, Informatics Society of Iran
- Apr 2020 **Semantic Storage and Retrieval Using the Resource Description Framework**, Shahid Beheshti University
- Dec 2019 **Simulating Self-Adaptive Software Architecture with the SWIM Scenario in the Rainbow Framework**, Faculty of Computer Science and Engineering, Shahid Beheshti University
- Dec 2017 **What is Net Neutrality, and Where Does it Come From?**, Sharif University of Technology
- Sep 2017 **A Comparative Introduction to MV* Patterns and Pattern Families**, Tehran Linux Users Group, Univ. of Tehran
- May 2016 **Mezzanine Content Management System Workshop (Pycon)**, Iran University of Science and Technology

HONORS & AWARDS

- May 2025 **Reviewer**, 41st International Conference on Logic Programming (ICLP 2025). [10.1017/S147106842510032X](#)
- Jul 2019 **Ranked 5th**, Nationwide Masters University Admission Exam among more than 15000 participants.
- Sep 2014 **23rd Team**, Sharif 5th Cyber Security and Hacking Competition (CTF) among over 800 teams. [Finals](#), [Quals](#)
- Nov 2013 **16th Team**, Sharif 4th Cyber Security and Hacking Competition (CTF) among over 100 teams. [Finals](#), [Quals](#)

WORK EXPERIENCE

- Nov 2022 **Technical Lead & DevOps**, [HUMMERS LEASING AND HUMMERS BROKER](#), Tehran, Iran
- Sep 2024
- Formed a high-performing technical team after systemically evaluating 500+ individuals in various areas.
 - Maintained communication with stakeholders to thoroughly grasp project requirements, progress, and risks.
 - Implemented automation scripts for DevOps processes, including email server and GitLab administration.
 - Implemented enterprise-level web software for leasing management using Django and Bootstrap.
- Team Leadership SDLC Nginx Redis Linux ATS Recruitment Django Zimbra Mail Server HTMX PostgreSQL GitLab CoCalc Moodle Selenium Agile Coaching
- Jul 2020 **Graduate Researcher, DATA-ORIENTED RESEARCH FOR SOFTWARE ANALYTICS LAB**, Tehran, Iran
- Oct 2022
- Developed two recommender systems to help library selection. One used recent deep learning algorithms, while the other used content-based recommenders to alleviate the cold-start problem.
 - Extracted GitHub data (< 3 mins) using BigQuery, filtered incorrect libraries based on Maven conventions, and built samples using repository info (stars, branches, authors, commits, lifetimes, etc.).
 - Extracted project topics and descriptions from GitHub API and processed them to complete tags.
 - Designed an interactive dashboard using Data Studio for seamless data exploration.
- Recommender System Deep Learning Python Google Cloud Google Colab BigQuery Data Studio Visual Paradigm
- Jul 2017 **Technical Lead & Web Developer**, [CONNECT ISTI](#), Tehran, Iran
- Sep 2024
- Developed web software using Django, implemented black box regression tests, administered Linux servers, and established a continuous delivery pipeline and data backup strategies.
 - Created detailed technical documents for various stakeholders and documented recruitment procedures.
 - Collected requirements, analyzed them, and divided them into actionable implementation tasks.
- Kanban System Analysis Software Architectural Design Django Semantic-UI PyCharm HTML5 CSS3 JavaScript
- Sep 2016 **Undergraduate Research Assistant, INTELLIGENT SOFTWARE ENGINEERING LAB**, Tehran, Iran
- Jul 2018
- Developed a JetBrains plugin to track user actions and a tool for detecting potential bugs by analyzing GitHub histories to find commits later fixed in subsequent revisions; acknowledged in "Cross-project code clones in GitHub" ([10.1007/s10664-018-9648-z](#)).
- GitHub Datasets Java Standard pygit2 gensim nltk GHTorrent code clone detection JGit JetBrains Plugin SQL